

Morphology controls of GeO₂ particles precipitated by a facile acid-induced decomposition of germanate ions in aqueous medium

Chengbin Jing, Jinxia Hou, Yongheng Zhang*

College of Materials Science and Engineering, Qingdao University of Science and Technology, 53 Zheng-zhou Road, Qingdao 266042, China

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Abstract

GeO₂ single crystals with various morphologies were synthesized at room temperature by an acid-induced homogenous liquid phase precipitation technique without using any surfactants as studied previously. With addition of aqueous ammonia, the solubility of hexagonal GeO₂ in water was significantly increased by formation of soluble germanate ions as confirmed by IR spectra analyses. Supersaturated GeO₂ solution could be produced by adding acid into the GeO₂-ammonia solution through the acid-induced transformation of germanate ions into GeO₂. GeO₂ spheres were obtained with phosphoric acid addition. Truncated cubic-like and cubic-like GeO₂ single crystals could be produced in the solutions with hydrochloric acid and ascorbic acid (vitamin C) additions, respectively. The morphology developments of the GeO₂ particles induced by various acids were discussed and the growth mechanism conforms to BFDH and HP crystal growth models.

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1. Introduction

Submicron or nano-particles often have novel physical and chemical properties, differing from those of the corresponding bulk materials [1–5]. For example, Submicron or nanometer-sized semiconductors have electronic and optical properties that depend on the particle size, making them potential candidates for applications, where tunability of these properties is essential [6–8]. The unique properties of submicron and nano-particles open up the general question of how crystallite size and shape affects the structural stability. The design and synthesis of exotic submicron or nano-particles have been investigated intensively in recent years but still remains a challenge. GeO₂ is a dielectric oxide and a blue photoluminescence material with peak energies around 2.2 and 3.1 eV, and it has a refractive index and a linear coefficient of thermal

expansion that are higher than those for SiO₂ [9–11]. Moreover, it is a promising material for optical waveguides and nano-connections in optoelectronic communication and vacuum technology [12–16]. Due to these fascinating properties, great attention has been paid to the preparation of GeO₂ nano-structures, including nano-particles, nano-fibers and nano-sheets [6–16].

The wet chemical process has been widely used to synthesize oxide particles, because the shapes and morphologies of the particles can be easily controlled by using a low-temperature growth process [3–5]. Germanium alkoxides or halides are the precursors generally used to synthesize GeO₂ materials through wet chemical process [12,14–20]. Because these precursors are pretty moisture-sensitive, the reversed micelles have often been used to obtain monodisperse GeO₂ particles [15,16]. Here, we report for the first time the synthesis of monodisperse GeO₂ with different shapes by an acid-induced homogenous liquid phase precipitation technique. The precipitation processes were carried out in surfactant-free aqueous

*Corresponding author. Fax: +0086 532 84022814.

E-mail address: yonghengzhang@yahoo.com (Y. Zhang).

solutions without using germanium alkoxides or halides as precursors. The simplicity and low cost of this work bring a great easiness to the industrialization of this technique. The chemical reactions and growth mechanism of GeO_2 particles were also investigated.

2. Experimental section

All chemicals (GeO_2 powders, aqueous ammonia (25 wt%), hydrochloric acid (36 wt%), phosphoric acid (85 wt%) and ascorbic acid (vitamin C, 99.9 wt%)) were supplied by Shanghai Aibi Chemistry Preparation Co. Ltd. and used as received without further purification.

Initially, 60 ml deionized water was mixed with 1.8 ml aqueous ammonia in a glass beaker followed by adding 1.38 g of GeO_2 powders. After stirring at room temperature for 40 min, the transparent solution was obtained. The as-synthesized solution was put into three beakers (each beaker contains 20 ml liquid sample) separately. The pH values of these solutions were adjusted to 3 with hydrochloric acid, phosphoric acid and ascorbic acid, respectively. After certain time, white precipitates were slowly produced in the solutions. The precipitation process was allowed to take place at room temperature, and then the precipitates were filtrated and ultrasonically washed in deionized water.

The infrared transmission spectra of the obtained particles were recorded using a Nicolet 60SXB IR spectrometer. The morphologies of the GeO_2 particles were observed by a scanning electron microscope (SEM, JSM-6700F) and the crystalline structure of the particles was analyzed by X-ray powder diffraction (XRD, X3000 Seifert diffractometer).

3. Results and discussions

Crystalline GeO_2 has mainly two crystal structures; one is hexagonal with a structure of low-quartz and the other is tetragonal with a structure of rutile [2,21–25]. Hexagonal GeO_2 is somewhat soluble in water (0.453 g per 100 g of water, 25 °C), while tetragonal GeO_2 is insoluble. Generally, the hexagonal GeO_2 exhibits a higher solubility at high pH value and the GeO_2 -aqueous ammonia has been used as the precursor solution for the preparation of GeO_2 -based sol–gel bulk materials [22,25]. In this work, it was found that the solubility of hexagonal GeO_2 could be increased to 2.3 g per 100 g of water by adding aqueous ammonia. The as-obtained solution was so stable in the ambient atmosphere that it could serve well as a precursor solution in this work.

It is agreed that the interaction between the hexagonal GeO_2 and ammonia can produce germanate complexes [2,15,22,23]. Fig. 1 shows the infrared spectra of original GeO_2 (a), ammonia solution (b) and GeO_2 solution containing ammonia (c). The strong absorption peak at 885 cm^{-1} in Fig. 1(a) is associated with the vibration mode ν_3 fundamental of the GeO_4 group [20]. It is well known

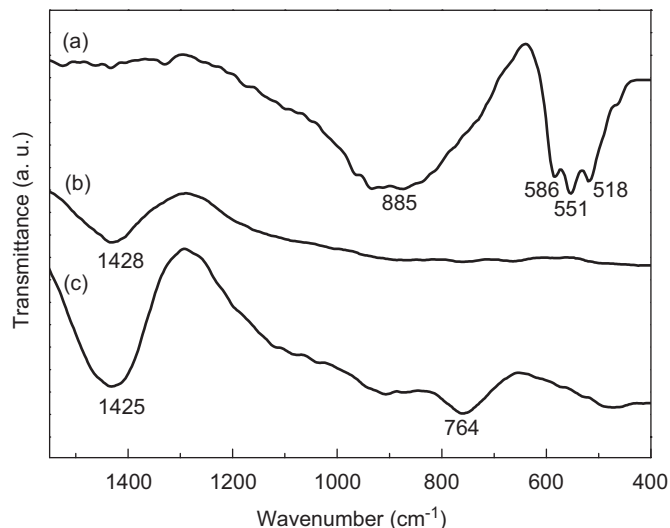


Fig. 1. The infrared spectra of original GeO_2 (a), ammonia solution (b) and GeO_2 solution containing ammonia (c).

that the hexagonal GeO_2 exhibits a typical triplet absorption peak within the wavenumber region from 587 to 516 cm^{-1} [24]. Such a triplet can be clearly observed in Fig. 1(a) confirming that the original GeO_2 has a hexagonal structure. Fig. 1(b) indicates that there is an absorption peak centered at 1428 cm^{-1} that can be assigned to the vibration of ammonium ions [26]. In Fig. 1(c), a new peak at 764 cm^{-1} was observed, which can be assigned to the Ge–O stretching modes of germinate ions [15,16,24]. There is no triplet absorption peak in the region of 587–516 cm^{-1} implying that GeO_2 had sufficiently reacted with ammonia. Furthermore, the intensity of the peak at 1425 cm^{-1} was increased with addition of GeO_2 indicating that the reaction between GeO_2 and ammonia produced more ammonium ions. In this work, the molar ratio between GeO_2 and ammonia was 1:1, and hence, the reaction between GeO_2 and ammonia probably is as follows:



Due to the formation of HGeO_3^- , a high amount of hexagonal GeO_2 could be dissolved in the aqueous ammonia.

When acid was designedly added to the as-synthesized transparent GeO_2 -aqueous ammonia, reaction (2) would proceed to the left side due to the decrease of OH^- . As a result, some HGeO_3^- would transform into GeO_2 leading to the increase of the GeO_2 solute concentration. The nucleation and growth of GeO_2 crystals in the solution started when the concentration of GeO_2 solutes was higher than its saturation solubility. According to the nucleation and growth kinetics of particles in a liquid medium, the nucleation rate u_1 and the growth rate u_2 follow Eq. (2) and

Table 1

The recorded time when precipitates were observed in the solutions after acid addition and the crystal sizes of the as-obtained GeO₂ particles

Acid category	Hydrochloric acid	Phosphoric acid	Ascorbic acid
Time when precipitates were observed after acid addition (h)	7	1	8
Crystallite size (after precipitation for 24 h)	568	89	714

Eq. (3), respectively [26] as

$$u_1 = k \left(\frac{C}{C_s} - 1 \right), \quad (3)$$

$$u_2 = \frac{D}{\delta} A (C - C_s), \quad (4)$$

where C is the solute concentration, C_s the normal solute saturation, k the constant, A the surface area of the particle, D the diffusion coefficient and δ the thickness of the superficial diffusion layer of the particle. Equations (3) and (4) indicate that the nucleation rate u_1 and the growth rate u_2 are functions of the solute concentration C . If the solute concentration is gradually increased, the nucleation and the growth of particles can go progressively resulting in a homogenous liquid-phase precipitation process. In this work, it was found that no white precipitates were immediately produced with addition of hydrochloric acid, phosphoric acid or ascorbic acid implying that the transformation of HGeO_3^- into GeO_2 solutes progressed very gradually. Various times were needed to start the precipitation process in these solutions (see Table 1). It was also found that the category of the acid has great influences on the shapes, sizes and morphologies of GeO_2 particles (Figs. 2–4). Truncated cube-like GeO_2 with a size of about 600 nm was produced in the solution with hydrochloric acid (see Fig. 2) after precipitation of 24 h. The obtained particles were single crystal with a hexagonal structure as confirmed by selected area electron diffraction (SAED) (shown in the insert of Fig. 2). However, the GeO_2 particles with two hexagonal pyramids at the ends and a hexagonal prism in the middle were observed as shown in Fig. 3 after precipitation for a shorter time of 7 h. When phosphoric acid was used as the induction agent, the spherical-like GeO_2 particles with a size of 50–150 nm were produced (Fig. 4). Interestingly, the monodispersed GeO_2 cubes with an average edge length of about 700 nm were formed in the solution containing ascorbic acid (Fig. 5) after 24 h of precipitation. The precipitated GeO_2 particles with various acids exhibit different appearances from that of the raw GeO_2 particles as shown in Fig. 6. X-ray powder diffraction (XRD) patterns of GeO_2 particles obtained at various conditions were shown in Fig. 7. The positions of all the diffraction

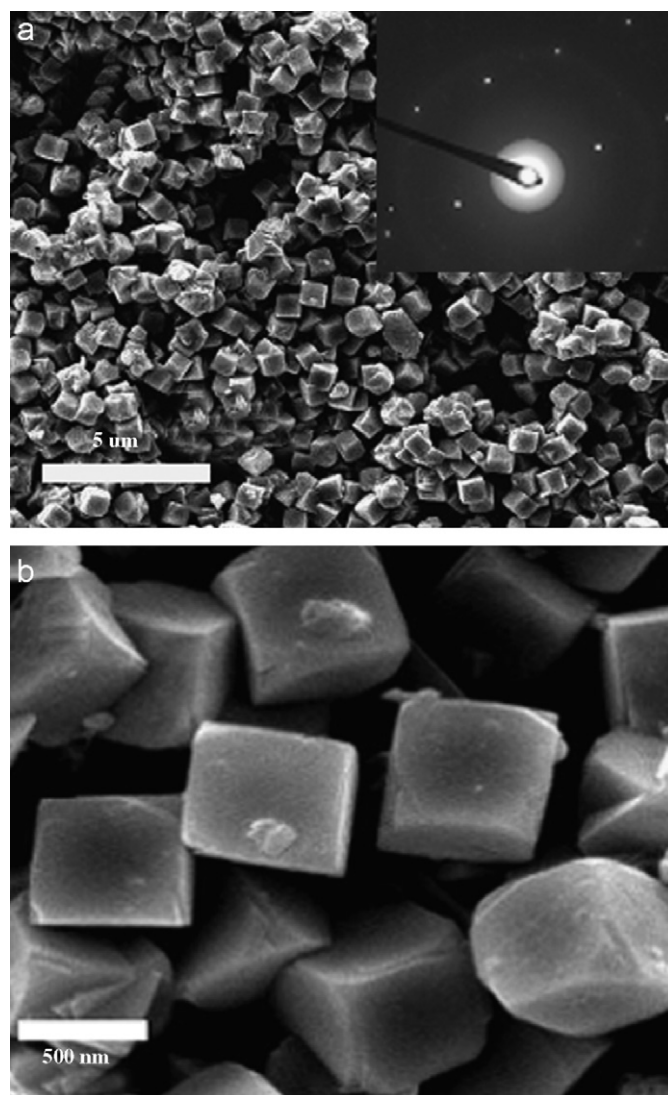


Fig. 2. The morphology of the particles precipitated by using hydrochloric acid after 24 h. (a) Low magnification and (b) high magnification view. The inset shows the selected area electron diffraction (SAED) pattern.

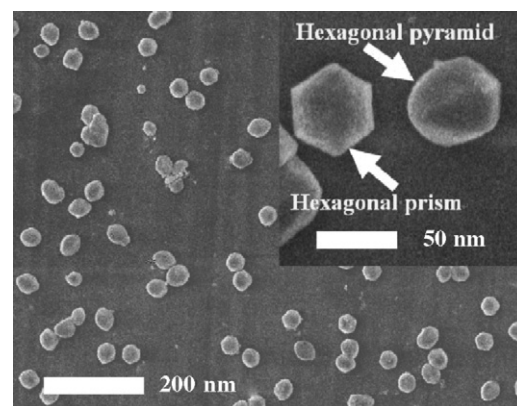


Fig. 3. The morphology of the particles precipitated by using hydrochloric acid after 7 h. The insert is a high magnification view.

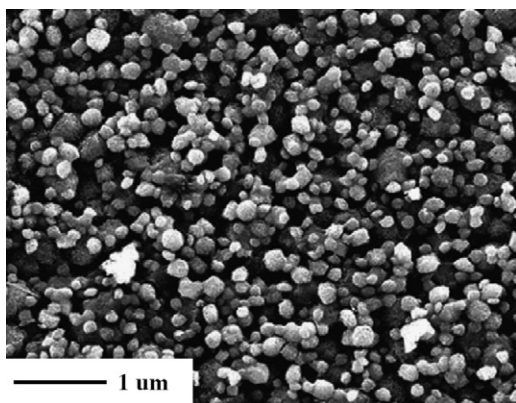


Fig. 4. The morphology of the particles precipitated by using phosphoric acid after 1 h.

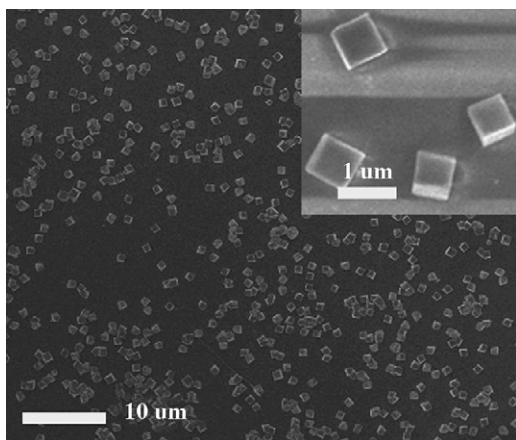


Fig. 5. The morphology of the particles precipitated by using ascorbic acid after 24 h. The inset shows the high magnification view.

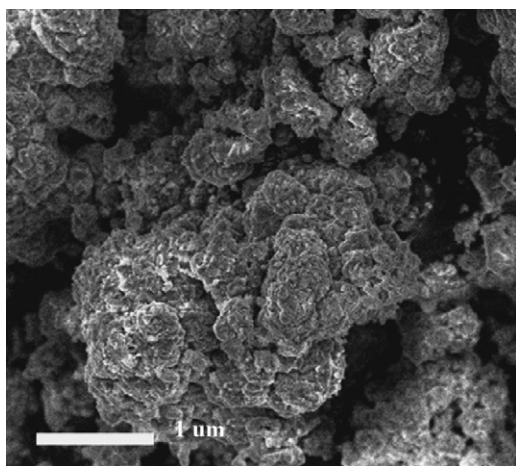


Fig. 6. The morphology of raw GeO_2 powders used for the synthesis of the GeO_2 -aqueous ammonia precursor solution.

peaks show a good agreement with those of the pure hexagonal GeO_2 (α -phase quartz-like structure, space group: $P3_112$) listed in the standard JCPDS card (36-1463) disclosing the overall crystallinity and purity of

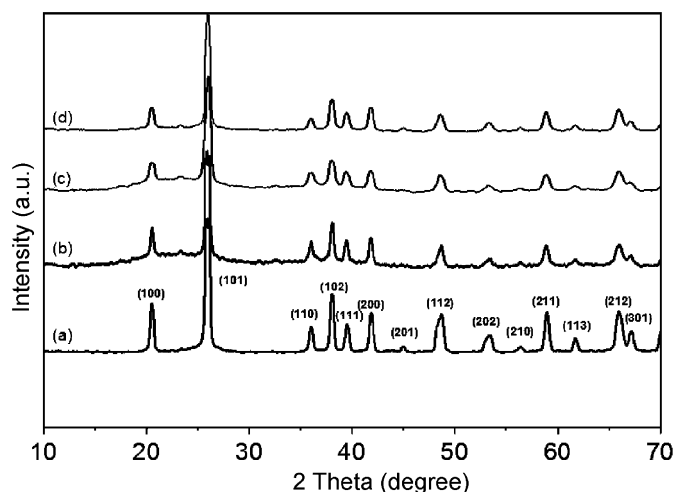


Fig. 7. The XRD patterns of raw GeO_2 powders (a) and the particles precipitated by using phosphoric acid (b), hydrochloric acid (c) and ascorbic acid (d).

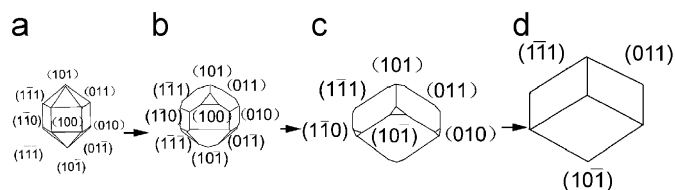


Fig. 8. Schematic of the geometries of GeO_2 particles during the growth process.

the as-synthesized GeO_2 particle products [20]. The crystallite sizes were calculated from the line broadening of the 101 diffraction peak ($2\theta = 26^\circ$) using the Scherrer's equation and the results were shown in Table 1, which agreed well with SEM observations. The GeO_2 particle samples collected from the GeO_2 -aqueous solution with phosphoric acid addition show the smallest crystallite size (89 nm) while the GeO_2 particles with largest crystallite size (714 nm) were produced in the solution containing hydrochloric acid.

The different morphologies of the precipitated GeO_2 crystals were attributed to the crystal growth mechanisms proposed by Bravais, Freidel, Donnay and Harker (BFDH model) and Hartman and Perdok (HP model) [15,16,27]. The BFDH model suggests that the slowest growing crystal planes are those with the highest atomic packing density and the largest spacing between adjacent layers. The planes having a strong tangential force and a weak normal force are the most dominant growth faces. The growth velocity along the normal direction of a crystal plane is inversely proportional to the interplanar spacing. For GeO_2 , the crystal planes $(1\bar{1}1)$, (101) , (011) , (010) , $(01\bar{1})$, $(10\bar{1})$, $(1\bar{1}\bar{1})$ and $(1\bar{1}0)$ with the highest atomic packing density [16] will grow slower compared with other planes resulting in the GeO_2 particles with a shape consisting of two hexagonal pyramids at the ends and a hexagonal prism in the middle as shown in Fig. 8a. With further growth of the

crystal, the HP model will significantly affect the evolution of crystal shape through the attachment energy of crystal plane as suggested by Hartman and Perdok that the faces with low attachment energies are the most prominent growth faces [27]. In GeO_2 , the crystal planes (011), (1 $\bar{1}$ 1) and (10 $\bar{1}$) of the hexagonal pyramids and hexagonal prism have high attachment energy [16], and therefore, they will grow more slowly resulting in the developments of spherical-like, truncated cube-like and finally perfect cubic-like shapes as schematically shown in Figs. 8b–d, respectively [16,27]. The evolution processes of the GeO_2 crystals during growth have been clearly confirmed by SEM observations.

Different morphologies of the crystals obtained by using different acids after certain precipitation time were due to their acidic differences and hence different stages of crystal growth stages. Hydrochloric acid and ascorbic acid carry one H^+ while phosphoric acid carries three H^+ and the ionization of these H^+ is possible with the progression of the chemical reactions in the GeO_2 -aqueous ammonia solution. So, phosphoric acid would supply more H^+ for the transformation of HGeO_3^+ into GeO_2 solutes than hydrochloric acid and ascorbic acid even the initial pH of the solution was the same. Therefore, after addition of phosphoric acid into the solution, the concentration of GeO_2 solutes in the solution will increase rapidly at the early period of time leading to the formation of a large number of GeO_2 nuclei, being followed by the growth process according to Eq. (2) and Eq. (3). This is why the precipitation of GeO_2 induced by phosphoric acid was the fastest (Table 1) and the particles were highly aggregated (Fig. 4). Theoretically, the cube-like shapes could also be developed using phosphoric acid according to BFDH and HP models. However, the high nucleation and growth rates would lead to a large consumption of the GeO_2 solutes in the solution within rather short time, and hence the growth rate of the GeO_2 particles would slow down rapidly limiting the particles to grow up into the cube-like shape. These GeO_2 particles could only grow up into the intermediate shapes between hexagonal pyramids and cubes looking like very small spheres (Fig. 4). The transformation of HGeO_3^+ into GeO_2 solutes induced by hydrochloric acid and ascorbic acid would result in the gradual formation of GeO_2 nuclei in smaller amount because of their less acidity. Therefore, the GeO_2 solutes were not largely consumed at the early period of the precipitation process allowing the existed GeO_2 nuclei to gradually grow up into the cubic-like shapes. Compared with hydrochloric acid, ascorbic acid is a weaker acid. Therefore, the release of H^+ should proceed mildly with the transformation process of germanate ions as demonstrated by the longest time required to generate the precipitates in the solution (Table 1). The continuous supply of H^+ would allow the as-formed GeO_2 nuclei to develop their shapes during a longer period of crystal growth leading to formation of GeO_2 particles with a perfect cubic shape and the largest

size of about 700 nm compared with the truncated cube-like particles induced by using hydrochloric acid.

4. Conclusions

The reaction between hexagonal GeO_2 and ammonia produced soluble germanate ions resulting in an increased solubility of GeO_2 in aqueous solution. Phosphoric acid, hydrochloric acid and ascorbic acid (vitamin C) were designedly added to the as-obtained transparent GeO_2 -aqueous ammonia solution to induce the transformation of germanate ions into GeO_2 solutes. However, the acidity of the acids would affect the nucleation and the growth rates of GeO_2 particles resulting in formation of final crystals with various morphologies after certain time. The spherical GeO_2 was obtained with the addition of phosphoric acid because of its high acidity, and hence, a faster nucleation and shorter time of crystal growth. The transformation of germanate ions induced by hydrochloric acid and ascorbic acid would result in the formation of GeO_2 particles with a truncated cubic-like and cubic-like shapes, respectively because of their weaker acidity. The crystal growth process induced by acid is in excellent agreement with the BFDH and HP models.

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